SASHWAT BILVESH DESAI

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EDUCATION

Master of Science in Applied Mathematics, Northeastern University, Boston 2024 - 2026
Probability, Mathematical Modeling, Applied Statistics, Machine Learning I & II, Analysis of Real Numbers.

Bachelor of Technology in Electronics Engineering 2019 - 2023
Dwarkadas Jivanlal Sanghvi College of Engineering, Mumbai, India

SKILLS

Programming Languages	Python, MATLAB, R, C++
Mathematical Skills	Differential Equations, Fourier Analysis, Linear Algebra, Calculus, Probability, Matrix Algebra, Vector Calculus, Statistical Inference, Numerical Methods, Graph Theory, Complex Analysis
Data Analysis Skills	Pandas, Matplotlib, PySpark, Plotly Dash, Data Assimilation Techniques, SciPy Ecosystem, Statistical Hypothesis Testing, Markov Chain Methods, Functional Data Analysis
Machine Learning Skills	Deep Learning, Regression, Classification, Optimization Techniques, Clustering, PyTorch, TensorFlow, OpenCV, Keras, Reinforcement Learning, Neural Networks, Natural Language Processing, Computer Vision, Transfer Learning
Soft Skills	Jupyter Notebooks, LaTeX, VS Code, GitHub, Mentoring, Grading, Reproducible Research Practices, Test-Driven Development, Experimental Design, High-Performance Computing, Documentation Tools, Research Paper Writing, Parallel Programming, Grant Proposal Preparation, Academic Assistance

EXPERIENCE

Data Club Leader Jan 2026 - Apr 2026
Northeastern University College of Science *Boston, MA*

- Led a weekly data community bridging coursework and industry via technical prep (LeetCode, SQL, statistics) and project showcases.
- Mentored members through resume reviews and interview debriefs, improving data science and analytics readiness.

Graduate Teaching Assistant Sep 2025 - Dec 2025
Northeastern University College of Science *Boston, MA*

- Held twice-weekly office hours on R and biological data analysis for MATH 7340: Statistics for Bioinformatics (Dr. Valerie Hower).
- Taught probability, distributions, inference, multiple hypothesis testing, ANOVA, and regression through problem-solving sessions.

Bridge To Calculus Summer Mentor Jun 2025 - Aug 2025
Northeastern University *Boston, MA*

- Mentored students in Calculus and Statistics and led R lab sessions on hands-on data analysis and statistical computing.
- Supported lead instructors with individualized academic help, grading, and attendance tracking.

Data Analyst and Machine Learning Research Assistant Mar 2024 - Aug 2024
Dwarkadas J. Sanghvi College of Engineering *Mumbai, India*

- Built a deep learning diabetic retinopathy detector with Prof. Shruti Dodani, comparing CNN, DenseNet, and ResNet architectures.
- Expanded the dataset from 102 to 703 fundus images using flipping, rotation, and brightness augmentation.

- Reached 97.17% test accuracy with DenseNet-169, using dropout and batch normalization to prevent overfitting.

PROJECTS

Property Tax Shift Research — Duke University Sponsored Project Jan 2026 - Apr 2026

- Trained Ridge, XGBoost, and LightGBM valuation models on 1.1M Cook County properties (60+ features; LightGBM $R^2 = 0.83$, MAPE = 29.4%) and designed a hybrid CCAO-ratio land value decomposition validated at a 20.8% median site-to-total ratio.
- Simulated revenue-neutral tax scenarios on 672,719 homeowners (57.5% pay less under a land-only tax, median savings $1,656), traced a spurious AUC = 1.0 to data leakage with SHAP, and extended to 5 metros using persistent homology (D.C.: 62.0% pay less; Philadelphia: 13.4%).

Barista Bench Feb 2026

- Benchmarked Gemini Flash Lite, GPT-4o-mini, and GPT-4.1 on 3,500 coffee orders for structured JSON extraction, adding a deterministic price validator that reached near-100% pricing accuracy.
- Engineered item deduplication and modifier normalization, scoring 0.76 exact match on the public leaderboard across 23 menu items, 5 sizes, and 20 modifiers.

Premier League Prediction Dec 2025 - Feb 2026

- Built a Selenium/BeautifulSoup FBRef scraper and a weighted ensemble (XGBoost, LightGBM, Random Forest, AdaBoost, Ridge) on 30+ features, with 10,000-run Monte Carlo simulation for title-race probabilities.
- Deployed as a FastAPI/Docker REST API on Render with weekly automated retraining via GitHub Actions.

Netflix Show Clustering Using K-Means Nov 2025

- Segmented 8,807 Netflix titles with K-means on 106 engineered features (TF-IDF, multi-label genres, one-hot encoding), choosing cluster count via Silhouette, Davies-Bouldin, and Calinski-Harabasz scores.
- Built a Euclidean-distance content recommender within clusters, achieving 97.17% clustering stability and separating international TV shows (30.4%) from mainstream movies (69.6%).

Shakespearean Text Generator Using Recurrent Neural Networks Nov 2025

- Implemented a 512-unit word-level LSTM on a 5.7MB Shakespeare corpus using 20-word sliding-window sequences and one-hot encoding.
- Engineered temperature-based multinomial sampling, generating coherent 600+ word passages in Shakespearean style after 100 training epochs.

Bach Chorale Generator Using Recurrent Neural Networks Nov 2025

- Implemented a 64-unit LSTM in TensorFlow/Keras on 382 JSB Chorales with 16-timestep sliding windows to model four-part harmony.
- Generated 100-timestep chorales via temperature-controlled sampling and synthesized audio at 22050 Hz; test loss of 0.0089 MSE and 0.0677 MAE.

Tic-Tac-Toe Using Reinforcement Learning May 2025

- Implemented a PyTorch Deep Q-Network with experience replay and decaying epsilon-greedy exploration, trained through self-play.
- Built the game environment (agent vs. agent, human vs. agent) and evaluated against random and minimax opponents on win and draw rates.

Impact of the COVID-19 Pandemic on Students' Behavior and Well-being Feb 2025 - Apr 2025

- Analyzed survey data from 1,182 students using chi-square tests, ANOVA, logistic regression, and correlation analysis.
- Found a negative correlation ($r = -0.16, p < 0.001$) between self-study and social media use, and identified physical activity and sleep as protective health factors.

Interconnected Economic Growth of Two Regions via Capital and Labor Flow Oct 2024 - Dec 2024

- Integrated the Solow Growth and Malthusian Population models into four differential equations linking capital, labor, and inter-regional mobility.

- Proved existence and stability of unique equilibria and identified a critical threshold in capital-induced labor movement affecting regional convergence.

Detection of Diabetic Retinopathy Using Deep Learning Mar 2024 - Aug 2024

- Built CNN, DenseNet-121/169, and ResNet-50 models on fundus images from Malpani Eye Hospital, labeled by a senior ophthalmologist.
- Achieved 97.17% test accuracy (DenseNet-169) on 703 augmented images and 93.75% on the raw 102-image dataset, using batch normalization and dropout.

Cost Optimization Solutions for E-commerce Vendors Apr 2022 - May 2023

- Built a unified cost model (material, production, transport, inventory) for perishable dairy e-commerce supply chains over a 4-layer vendor-to-customer network.
- Solved it with Differential Evolution in Python/NumPy on real dairy cost data, validated on the Rosenbrock function ($f = 0.03$ within 50 iterations).

PUBLICATIONS

- S. Desai and S. Dodani, "Detection of Diabetic Retinopathy Using Deep Learning Techniques," 2024 IEEE 3rd World Conference on Applied Intelligence and Computing (AIC), Gwalior, India, 2024, pp. 898-904.
- Sashwat Desai, "Cost Optimization Solutions for E-Commerce Vendors," International Journal of Innovative Science and Research Technology (IJISRT), Volume 8 Issue 9, September 2023, pp. 1317-1337
- Desai Sashwat Bilvesh, "Image Processing Techniques: A Review," International Journal of Innovative Science and Research Technology (IJISRT), Volume 8 Issue 2, February 2023, pp. 1302-1308